

Ilera-ATLAS: validation evidence

NAIC 2026 · Academia & Research Track · Problem Statement 3 (Sectoral Fine-Tuning) · October 2026 · github.com/OkeyAmy/Ilera

1. What was validated

Ilera-ATLAS (QLoRA fine-tune of NCAIR1/N-ATLaS on the 2024 National Standing Orders) against the raw N-ATLaS model, on **Ilera-Bench**, a benchmark frozen and hashed **before training**, plus an external multiple-choice test.

Set	Items
English vignettes and Q&A	196
Hausa (parallel)	150
Yoruba (parallel)	150
Unseen-section probe	9
Frozen total	505
External: AfriMed-QA multiple choice (test split)	300

Benchmark hash (bench/BENCH_SHA256.txt): 76e00a1adfff30db74f8e136343518210b828ecc93a16fc6a8cf2e2c1705c51c
ilera_bench.jsonl c467e9517a722989ffb0313501d8a99297eb6214ba300b9adf157c76ea64a42f afrimedqa_ids.txt

Leak control: no benchmark item, translation of one, probe-section text or near-duplicate (RapidFuzz ≥ 85) enters training. Statistics: paired bootstrap 95% confidence intervals against the raw model.

Baseline: raw N-ATLaS before fine-tuning

NCAIR1/N-ATLaS loaded **as released** (float16, no quantisation, no fine-tuning) on a Kaggle T4 x2, asked to find the Standing Orders reference for every part of the CHEW document ([01b_base_reference_eval](#), 6 Oct 2026; data in [results/](#)):

Test	Accuracy [95% CI]	Chance
Closed book: section number for a topic (105 items)	1.0% [0.0, 2.9]	0.7%
Excerpt + table of contents → section number (165 items)	9.1% [4.8, 13.3] (right title 17.0%)	0.7%
Excerpt → right section among 4 options (165 items)	75.2% [68.5, 81.8]	25%

The base model understands Standing Orders content well, but cannot produce the correct reference on its own, even with the table of contents in front of it. Ilera-ATLAS therefore (1) trains on answers that always end with the exact SOURCE: section and (2) is evaluated with the cited section as a metric (source_ok), with retrieval of the matching chunk as the planned stretch.

Results: Ilera-ATLAS vs raw N-ATLaS (frozen Ilera-Bench)

Chosen model: `ilera-atlas-lora-runB` (QLoRA $r=16$, 2 epochs, 1,058 examples; picked over the 3-epoch run on the validation set for its lower under-referral). Benchmark frozen before training (SHA-256 in [bench/](#)). Paired bootstrap, 95% CI; all differences below are significant.

Metric (496 items: 196 en + 150 ha + 150 yo)	Raw N-ATLaS	Ilera-ATLAS	Difference [95% CI]
Correct action (TREAT / REFER / URGENT)	1.8%	65.3%	+63.5 [+59.5, +67.7]
Correct action, lenient (action inferred from free text)	26.4%	65.3%	+38.9 [+33.3, +44.8]
Under-referral (dangerous error), strict	98.2%	18.5%	-79.6 [-83.3, -76.2]
Under-referral, lenient	55.8%	18.5%	-37.3 [-42.5, -32.5]
Answer format valid	8.3%	99.6%	+91.3
Key-fact recall	5.9%	23.0%	+17.1 [+14.7, +19.6]
Cites the correct Standing Orders section	0.0%	24.6%	+24.6 [+21.0, +28.4]
Answers in the question's language	80.8% (Yoruba 47.3%)	100%	+19.2

Per language, correct action: English 66.3%, Hausa 69.3%, Yoruba 60.0%. Unseen-section probe (9 items): 77.8% vs 0%. External AfriMed-QA multiple choice (300, general medicine): 38.0% vs 41.7% base, a small, non-significant drop: the model specialises in the Standing Orders. Full tables in [results/](#).

Limitations. Small training set (1,113 examples, AI-assisted review), English benchmark below the planned 300 items, AI-assisted (not yet fully clinician-verified) review of data and translations, and a blind clinical review of 150 answer pairs pending. The model can produce clinically inappropriate steps (e.g. IV medicines outside CHEW scope): it is decision support for trained workers only.

4. Full comparison table ([results/compare_base_vs_ilera.csv](#))

Metrics: `action_ok` (strict), `*_lenient` (action inferred from free text), `under_referral` (gold REFER/URGENT answered as lower; the dangerous error), `over_referral`, `keyfact` (recall), `source_ok` (correct section), `format_ok`, `lang_ok` (GlottLID).

Subset	Lang	Metric	Raw N-ATLaS	Ilera-ATLAS	Diff [95% CI]	n	Sig.
main	en	action_ok	4.1%	66.3%	+62.2 [+55.1, +69.9]	196	yes
main	en	under_referral	95.9%	17.3%	-78.6 [-84.2, -73.0]	196	yes
main	en	over_referral	0.0%	16.3%	+16.3 [+11.2, +21.9]	196	yes
main	en	action_ok_lenient	31.6%	66.3%	+34.7 [+25.5, +43.9]	196	yes
main	en	under_referral_lenient	42.4%	17.3%	-25.0 [-32.6, -17.3]	196	yes
main	en	format_ok	19.4%	100.0%	+80.6 [+75.0, +86.2]	196	yes
main	en	key_fact_recall	13.6%	27.6%	+14.0 [+9.2, +18.9]	196	yes
main	en	source_ok	0.0%	26.0%	+26.0 [+20.4, +32.6]	196	yes
main	en	same_language	100.0%	100.0%	+0.0 [+0.0, +0.0]	195	no
main	ha	action_ok	0.0%	69.3%	+69.3 [+62.0, +76.7]	150	yes
main	ha	under_referral	100.0%	16.7%	-83.3 [-89.3, -77.3]	150	yes

Subset	Lang	Metric	Raw N-ATLaS	Ilara-ATLAS	Diff [95% CI]	n	Sig.
main	ha	over_referral	0.0%	14.0%	+14.0 [+8.7, +19.3]	150	yes
main	ha	action_ok_lenient	22.0%	69.3%	+47.3 [+38.0, +56.0]	150	yes
main	ha	under_referral_lenient	67.3%	16.7%	-50.7 [-58.7, -42.7]	150	yes
main	ha	format_ok	0.0%	99.3%	+99.3 [+98.0, +100.0]	150	yes
main	ha	key_fact_recall	0.8%	18.3%	+17.6 [+13.9, +21.6]	150	yes
main	ha	source_ok	0.0%	26.7%	+26.7 [+19.3, +34.0]	150	yes
main	ha	same_language	89.3%	100.0%	+10.7 [+6.0, +15.3]	150	yes
main	yo	action_ok	0.7%	60.0%	+59.3 [+52.0, +67.3]	150	yes
main	yo	under_referral	99.3%	22.0%	-77.3 [-84.0, -70.7]	150	yes
main	yo	over_referral	0.0%	18.0%	+18.0 [+12.0, +24.7]	150	yes
main	yo	action_ok_lenient	24.0%	60.0%	+36.0 [+26.0, +45.3]	150	yes
main	yo	under_referral_lenient	62.0%	22.0%	-40.0 [-49.3, -30.7]	150	yes
main	yo	format_ok	2.0%	99.3%	+97.3 [+94.7, +99.3]	150	yes
main	yo	key_fact_recall	0.9%	21.8%	+20.8 [+17.3, +24.6]	150	yes
main	yo	source_ok	0.0%	20.7%	+20.7 [+14.7, +27.3]	150	yes
main	yo	same_language	47.3%	100.0%	+52.7 [+44.7, +60.7]	150	yes
probe	en	action_ok	0.0%	77.8%	+77.8 [+44.4, +100.0]	9	yes
probe	en	under_referral	100.0%	11.1%	-88.9 [-100.0, -66.7]	9	yes
probe	en	over_referral	0.0%	11.1%	+11.1 [+0.0, +33.3]	9	no
probe	en	action_ok_lenient	0.0%	77.8%	+77.8 [+44.4, +100.0]	9	yes
probe	en	under_referral_lenient	77.8%	11.1%	-66.7 [-88.9, -33.3]	9	yes
probe	en	format_ok	0.0%	100.0%	+100.0 [+100.0, +100.0]	9	yes
probe	en	key_fact_recall	11.1%	33.3%	+22.2 [+8.3, +38.9]	9	yes
probe	en	source_ok	0.0%	0.0%	+0.0 [+0.0, +0.0]	9	no
probe	en	same_language	100.0%	100.0%	+0.0 [+0.0, +0.0]	9	no
main	ALL	action_ok	1.8%	65.3%	+63.5 [+59.5, +67.7]	496	yes
main	ALL	under_referral	98.2%	18.6%	-79.6 [-83.3, -76.2]	496	yes
main	ALL	over_referral	0.0%	16.1%	+16.1 [+12.9, +19.2]	496	yes
main	ALL	action_ok_lenient	26.4%	65.3%	+38.9 [+33.3, +44.8]	496	yes
main	ALL	under_referral_lenient	55.9%	18.6%	-37.3 [-42.5, -32.5]	496	yes
main	ALL	format_ok	8.3%	99.6%	+91.3 [+88.9, +93.8]	496	yes
main	ALL	key_fact_recall	5.9%	23.0%	+17.1 [+14.7, +19.6]	496	yes
main	ALL	source_ok	0.0%	24.6%	+24.6 [+21.0, +28.4]	496	yes
main	ALL	same_language	80.8%	100.0%	+19.2 [+15.8, +22.8]	495	yes

5. Live system test (8 October 2026)

Deployed demo (huggingface.co/spaces/Emmanuel-okoye/ilera-atlas; Q4_K_M GGUF on Kaggle CPU) tested with three cases, recorded in the demonstration video:

Case	Language	Result
2-year-old, fever 3 days, fast breathing	English	TREAT, age-banded doses, refer if no improvement, cites Section 2.5
6-month-old, fever and cough 2 days	Yoruba	TREAT, four steps entirely in Yoruba
Same child as case 1	Hausa	URGENT, pre-referral ceftriaxone, cites Section 2.3
1-year-old with convulsions and fever	English	Safety layer triggered: danger-sign banner, URGENT

Observation: cases 1 and 3 show the action can differ across languages for the same patient; reported as a limitation. A blind clinical review of 150 base-vs-llera answer pairs is in progress.